

FINAL EXAMINATION SEMESTER 1 2014/2015

SUBJECT CODE : SCSR2033

SUBJECT : COMPUTER ORGANIZATION AND ARCHITECTURE

DURATION : 3 HOURS

DATE :

PLACE :

INSTRUCTIONS :

ANSWER ALL QUESTIONS FOR PART A AND PART B IN THE QUESTION BOOKLET.

(Please fill in your particulars)

Name								
Matric No.								
Years / Course								
Section								
Lecturer's Name (Please circle)	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; height: 20px;"></td> <td style="width: 50%; height: 20px;"></td> </tr> <tr> <td style="height: 20px;"></td> <td style="height: 20px;"></td> </tr> <tr> <td style="height: 20px;"></td> <td style="height: 20px;"></td> </tr> </table>							

This questions paper consists of 8 printed pages, excluding this page.

ANSWER ALL QUESTIONS.

THE QUESTION MUST BE WRITEN IN ANSWER BOOKLET.

Part A: Objectives (20 Marks)

1. There are three type of memory. Which of these is NOT a type of memory?
 - A. Main Memory
 - B. Light Memory
 - C. Secondary Memory
 - D. Cache Memory
2. The following statements are correct EXCEPT:
 - A. A hit is when data is found at a given memory level.
 - B. A miss is when data is not found at a given memory level
 - C. The hit rate is the average of time data is found at a given memory level.
 - D. The hit time is the time required to access data at a given memory level.
3. This volatile memory is commonly used as main memory.
 - A. EPROM
 - B. SRAM
 - C. DRAM
 - D. SDRAM
4. Which of following is NOT a replacement algorithm for cache memory?
 - A. Least frequently used (LFU)
 - B. First in first out (FIFO)
 - C. Least recently used (LRU)
 - D. Last in first out (LIFO)
5. Select a statement that is INCORRECT about virtual memory.
 - A. Virtual memory and main memory are divided equal sized pages.
 - B. Virtual memory pages allocated to a process must be stored contiguously.
 - C. Virtual memory maps page frames to virtual pages.
 - D. Virtual memory uses internal disk storage like hard disk.
6. Choose the BEST statement to describe Cache memory.
 - A. Inexpensive in cost with large capacity of location.
 - B. Usually located near main memory.
 - C. It keeps a copy of the most frequently used data from the main memory.
 - D. It sits at the top of the memory hierarchy.
7. The following statements are related to which data storage.

- Disk tracks are numbered from the outside edge, starting with zero.
 - Hard disk platters are mounted on spindles.
 - Read/write heads are mounted on a comb that swings radially to read the disk.

 - A. Magnetic Tape
 - B. Optical Disks
 - C. Magnetic Disk Technology
 - D. RAID

8. The following are optical disks EXCEPT:
- A. CDROM
 - B. Floppy disk
 - C. DVD
 - D. Blu-Ray
9. Which dominant re-coding method is used for magnetic tape?
- A. Up Down scan
 - B. Horizontal scan
 - C. Helical scan
 - D. Vertical scan
10. Which data storage has the following characteristics?

Data is stored across many disks
Has 0-6 level types

- A. Floppy disk
 - B. DVD
 - C. CDROM
 - D. RAID
11. Which of these is a type of I/O devices?
- A. Integer I/O devices
 - B. Block I/O devices
 - C. Bit I/O devices
 - D. Stream I/O devices
12. If **AX = 2345H** and **BX = 1BCDH**, what is the result of **AND AX, BX**?
- A. AX = 3BCDH
 - B. AX = 0345H
 - C. AX = 3FCFH
 - D. AX = 3888H
13. Study the assembly code given. What is the final result of register CH?

```
MOV AH, 3
MOV BH, 14
MOV CH, 7
ADD AH, BH
JC HERE
MOV CH, 1
JMP QUIT
HERE:
MOV CH, 5
QUIT:
```

- A. 1
- B. 5
- C. 7
- D. 0

14. You are given the assembly code below. Please identify if the line indicated is a valid code.

```
.DATA
SUM WORD 126
TOTAL WORD 55H

.CODE
MAIN PROC
MOV AX, BH      ;LINE1
ADD SUM, TOTAL  ;LINE2
SUB AX, SUM      ;LINE3
```

- A. LINE1 = valid; LINE2 = invalid; LINE3 = valid
B. LINE1 = invalid; LINE2 = invalid; LINE3 = valid
C. LINE1 = invalid; LINE2 = invalid; LINE3 = invalid
D. LINE1 = valid; LINE2 = valid; LINE3 = valid
15. Study the assembly code given below. If the first user input is 12 and the second is 8, what is the correct output?

```
.DATA
STR1 BYTE "ENTER A NUMBER :",0
.CODE
MAIN PROC
MOV EDX, OFFSET STR1
CALL WRITESTRING
CALL READDEC
MOV EDX, OFFSET STR1
CALL WRITESTRING
CALL READDEC
CALL WRITEDEC
CALL WRITEDEC
```

- A. 12 8
B. 12 12
C. 8 12
D. 8 8
16. Study the assembly code given below. Choose the correct output.

```
.data
str1 byte "Test1 ",0
str2 byte "Test2 ",0
trest byte "aBc ", 0
str3 byte "Test3 ",0

.code
main PROC
mov edx,offset trest

mov eax, offset str3
call writestring
```

```

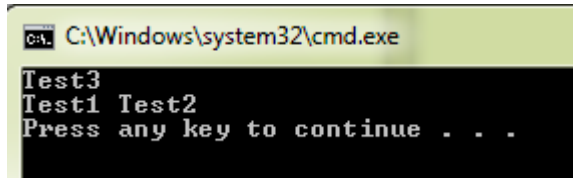
call crlf

mov edx, offset str1
call writestring

mov edx, offset str2
call writestring
call crlf

```

A.

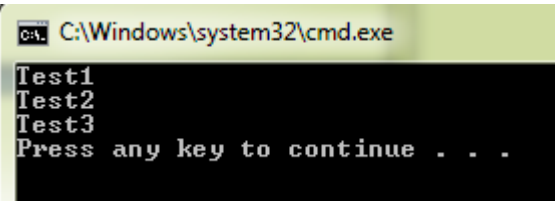


```

C:\Windows\system32\cmd.exe
Test3
Test1 Test2
Press any key to continue . . .

```

C.

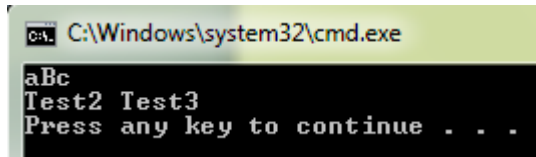


```

C:\Windows\system32\cmd.exe
Test1
Test2
Test3
Press any key to continue . . .

```

B.

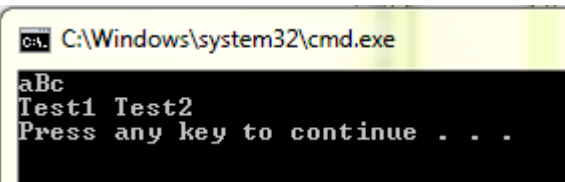


```

C:\Windows\system32\cmd.exe
aBc
Test2 Test3
Press any key to continue . . .

```

D.



```

C:\Windows\system32\cmd.exe
aBc
Test1 Test2
Press any key to continue . . .

```

17. Which of the following is NOT a measure for computer performance?

- A. Response time
- B. RAM size
- C. Throughput
- D. Execution time

18. All the followings affect CPU performance EXCEPT:

- A. Programming language
- B. Printer speed (ppm)
- C. ISA
- D. Compiler

19. Your friend is a normal user of computer; what would be the most important feature for him?

- A. RAM size
- B. Response Time
- C. Execution time
- D. Throughput

20. In comparing computer systems, to say that System A is faster than System B because System A runs at higher clock rate than system B is valid ONLY when:

- A. The ISAs of Systems A and B are different
- B. The ISAs of Systems B and A are identical
- C. The Printers used in System A and B are identical
- D. The Programs installed in Systems A and B are identical

Part B (80 Marks)

Question 1 (19marks)

1. Main memory is divided into blocks. If the memory word is 16 bit and the size of a block is 8 words.
 - a. What is the capacity of the main memory, if the total number of blocks in the memory is 1024? [3]
 - b. How many blocks in the main memory if the memory capacity is 64 Kbit? [3]
2. Given a memory address as 56Ah (11 bits) with 8 memory banks
 - a. How many bits are needed to address the memory banks. [2]
 - b. Draw the LOI (low order interleaving) address format. [2]
 - c. Calculate the memory bank capacity. [2]
 - d. Which module/bank that addresses 56Ah belongs to? [2]
 - e. Draw and label the sequence of address for the first and last module/bank using LOI. [5]

Question 2 (11marks)

1. In *Associative Mapping*, given the capacity of the cache memory is 32 MWords and the main memory is double of that size. Assume that 1 word = 16 bits.
 - a. Find the number of bits for main memory address and the cache address. [2]
 - b. Each line/location in the cache memory stores a block of 8 words. Determine the number of bits to address the words and find the size of a block in bits. [2]
 - c. Draw the format of main memory address. [2]
 - d. Find the cache address that mapped to the main memory address 3456789h. [2]
2. A CPU has cache's hit rate of 0.65 and the hit time is 3 cycles. If a miss occurred, the miss penalty is three times slower than the hit time. Given a cycle is 0.02 milliseconds, calculate the Average Memory Access Time (AMAT) in cycles and seconds.[3]

Question 3 (15marks)

1. State TRUE/FALSE for the following statements. [6]

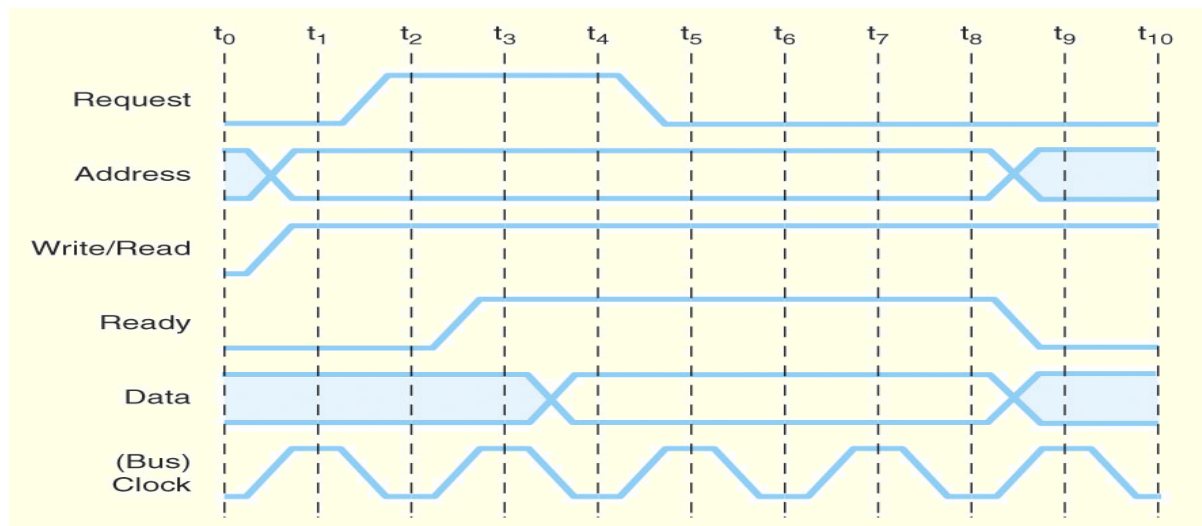
Statements	TRUE/FALSE
a. I/O performance is measured based on data rate (throughput) and latency.	
b. Interrupt I/O is also known as polling.	
c. For Interrupt I/O transfer control, system state is saved before the interrupt service routine is executed and is restored afterward.	
d. For DMA transfer control, CPU offloads execution of I/O instructions to DMA.	
e. DMA consists of one or more I/O processors (IOPs) that control various channel paths.	
f. Programmed and Interrupt I/O is CPU controlled. All I/O goes through the CPU.	

2. The following parameters are used for processor overhead calculation for interrupt I/O and DMA. [5]

- 1000 MHz CPU
- Interrupt handler takes 400 cycles
- Data transfer takes 100 cycles
- 8 MB/s, 16 B interface disk transfers data 50% of time
- DMA setup takes 800 cycles, transfer one 16KB page at a time

- Calculate processor overhead for interrupt I/O
- Calculate processor overhead for DMA

3. Based on the timing diagram, complete the following table: [4]



Time	Bus Signal	Meaning
t ₀		
t ₁		
t ₂		
t ₃		

Question 4 (15marks)

1. Consider two different machines, with two different instruction sets, both of which have a clock rate of 200 MHz. The following measurements are recorded on the two machines running a given set of benchmark programs.

Instruction Type	Instruction Count (millions)	Cycles Per Instruction (CPI)
Machine A		
Arithmetic and Logic	8	1
Load and Store	4	3
Branch	2	4
Others	4	3
Machine B		
Arithmetic and Logic	10	1
Load and Store	8	2
Branch	2	4
Others	4	3

- a) Determine the CPU Clock Cycles for each machine. [2]
 - b) What is the CPI for each machine? [3]
 - c) Determine CPU execution time for each machine. [3]
 - d) Comment on the results. [2]
2. Assume a program runs in **25** seconds on computer **A**, which has **12 GHz**. You are trying to help your friend who is a computer designer to build computer **B** that will run the same program in **5** seconds. Your friend has determined that increasing the clock rate is possible; however, he also realized that this will affect the rest of the CPU design, causing computer **B** to require 2.5 times as many clock cycles as computer **A** for this program. What clock rate should you advise your friend to target for? [5]

Question 5 (20marks)

1. Given the array definition below, write the appropriate assembly code to sum the array.
[5]

NUMBS DWORD 4, 15, 16, 24, 26

2. Given the pseudocode below, write the appropriate assembly code to achieve it.[5]

If X > 40 OR Y < 50 THEN JOJO = JOJO + X ELSE JOJO = JOJO + Y
--

3. Given the pseudocode below, write the appropriate assembly code to achieve it.[5]

If X is divisible by 4 then UPPER = UPPER + 4 ELSE OTHER = OTHER + 1

4. Given the assembly code instructions below, fill in the blanks with the correct answer.
Write your answer in hexadecimal. [5]

a.	MOV AX, 35CH AND AX, 444H ; AX = _____
b.	MOV BH, 45H OR BH, 00111100B ; BH = _____
c.	MOV CL, 23H XOR CL, 23H ; CL = _____
d.	MOV DL, 0ACH NOT DL ; DL = _____
e.	MOV AL, 5H MUL AL ; AX = _____